

Packager Guidelines

Section 5: Compressor Crankcase Oil System

5-1. Frame Oil Recommendations

General Requirements

The oil must be well-refined petroleum product with low carbon residue. It should not contain fats or fixed oil compounding. Oil containing rust and oxidation inhibitors and a foam depressant is highly recommended. The oil selected must be substantially non-corrosive to the common bearing metals, with a pour point at least 10°F below ambient temperature when the unit is started for engine drives. The Compressor Cylinder and Packing Lubrication System on new units is designed at the factory based upon the specific application. In most cases, the system is designed to draw lubricating oil from the compressor crankcase to supply the lubricator pumps unless there are specific reasons why a different type lubricating oil is required for the cylinders. Packager should define on the Distributor Order Form where they wish to obtain the oil supply to the lubricator. (See Compressor Cylinder and Packing Lubrication Guidelines.)

Multiviscosity lubricants may be used in the crankcase where widely variable ambient temperatures are encountered, provided that the lubricant maintains a viscosity of 115 to 150 SSU (23.7 to 31.8 cSt) at normal operating sump temperature. Provision must be made to heat the crankcase oil to minimum starting temperatures as outlined in 5-5 Oil Parameter Recommendations.

It is recognized that the oil recommended by the engine manufacturer (when an engine drive is supplied) is normally heavier (in the SAE 40 viscosity range). When this is the case and the user desires to stock only one oil for use in both the engine and compressor crankcases, the heavier weight oil recommended for the engine may also be used in the compressor crankcase. Under cold ambient starting conditions, it is recommended that the oil sump temperature be at least 10°F (5.6°C) to 20°F (2.8°C) above the pour point prior to starting the engine drives. This may require the use of auxiliary crankcase heaters.

If sump oil temperatures are 5°F (2.8°C) higher than the allowable maximum sump temperature for a specific viscosity, the lubricant should be monitored more closely for nitration effects and more frequent oil changes should be scheduled to prevent an increase in long term maintenance programs.

Viscosity Requirements

Table 2-1. Crankcase Oil Viscosity Requirements

	SUMP OIL TEMPERATURE		
	Below 145°F (68°C) SAE 20	145-165°F (63-74°C) SAE 30	165-180°F (74-92°C) SAE 40
VISCOSITY AT 100°F Saybolt Universal, SSU Kinematic Viscosity, cSt	420 90.5	600 129.5	750 162.5
VISCOSITY AT 210°F Saybolt Universal, SSU Kinematic Viscosity, cSt	50 7.3	60 10.5	70 13

Inspection

Conduct all testing of the lubricating oil in accordance with the Standard Methods (latest edition) of the American Society for Testing Materials (ASTM).

Oil Change Schedule

The oil should be changed every **2000 to 4000 hours**, or **every 6 months**, depending on local conditions, or as indicated by oil analysis. Most reputable oil companies offer laboratory analysis services of oil samples on request. We highly recommend the use of this type service. A once-a-month analysis schedule is recommended when the compressor is operated under severe conditions.

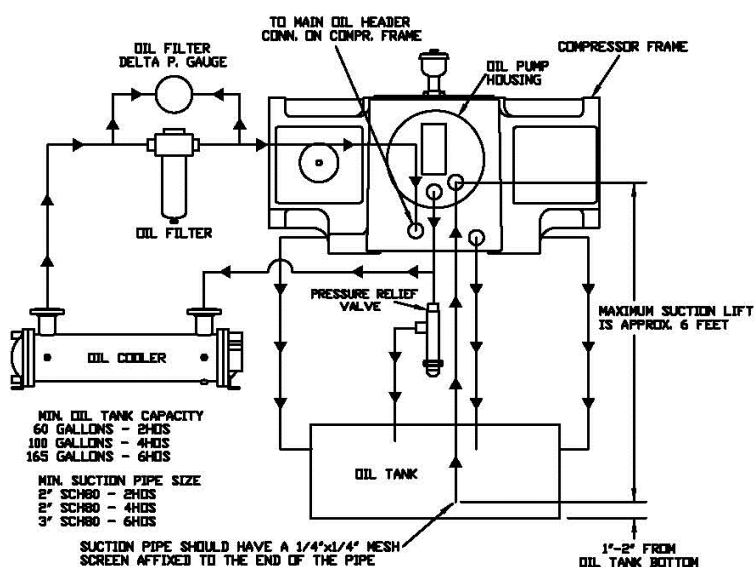
Where the compressor is operated in an extremely dirty atmosphere, installed outdoors, operated intermittently, handling foul gas or where the oil reaches high temperatures in a very hot atmosphere, it may be necessary to change the oil more often. In case of dirty atmosphere, consider using sweeping nitrogen purge in crankcase.

Dry Sump Requirements

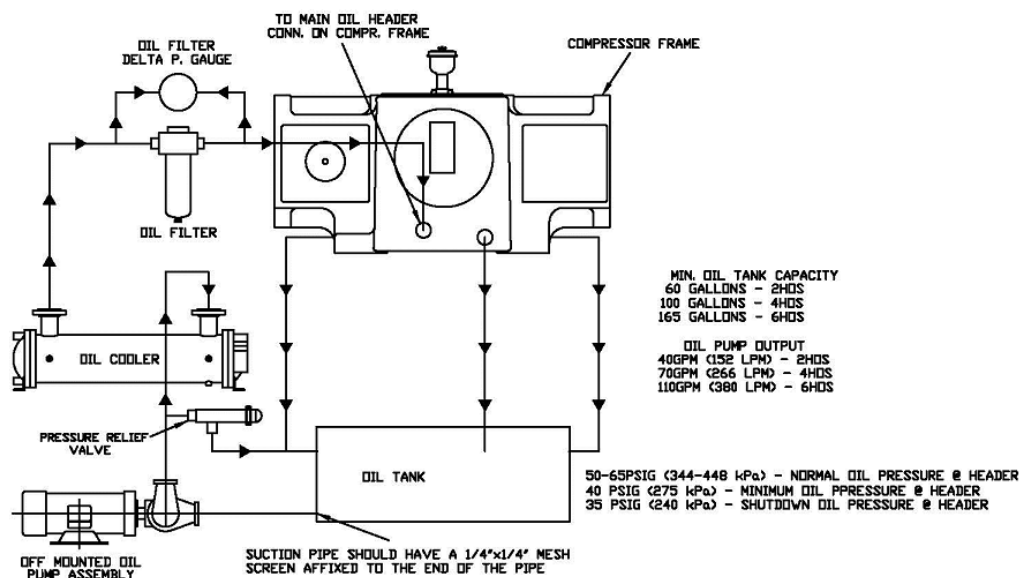
Any requirements outside these limitations require an auxiliary oil pump and a deep independent oil sump within the package.

Frame Size	Max. Pitch	Max Roll
MOS, HOS2	3.0°	13°
MOS, HOS4	1.5°	13°
MOS, HOS6	1.0°	13°
HOSS2	2.0°	11°
HOSS4	1.2°	11°
HOSS6	0.8°	11°

Dry Sump With Frame Mounted Pump



Dry Sump With Off Mounted Pump



5-2. Oil Cooler and Filter

All units have frame mounted oil filters that are installed at the factory and should not be removed for any reason. These filters are normally mounted off the oil pump end of the compressor; alternate locations may be used to suit a particular installation. Filter internals should be checked for cleanliness and correct positioning. Piping connections to the cooler and filter must line up properly before they are tightened. Do not spring the piping to make connections, as this may cause leaking joints or result in fatigue failures.

Some customers require duplex type oil filters. In these instances, Siemens Energy recommends the original frame mounted filter be left in place. After 200 hours of run time, the element should be removed and continue flowing through the empty filter housing. This ensures the oil system has removed any contaminants which may have been introduced during the packaging of the unit.

5-3. Oil Piping

The oil piping in the frame lubrication system downstream of the cooler to the filter must be closely examined and cleaned before charging the system with oil.

Where the customer is supplying external oil piping, it is his responsibility to provide piping that has been thoroughly cleaned and completely examined throughout. Any foreign material loosened by the flow of oil through the piping can seriously damage the compressor. All external oil piping that is supplied by the customer should be the same size as the connection on the frame. The piping size should NEVER be stepped down for any reason.

Any check valves included in the oil system should be flapper type FULL flow check valves. A piston check should NEVER be used since this can act as a restriction. Any type restriction can cause severe damage to the frame internals.

5-4. Prelube Pumps

Prelube pumps are required on frames that are driven by an electric motor or whenever the compressor may start-up while unattended. These prelube pumps may be electric or pneumatic driven, but require a permissive which will disable the start-up sequence if oil pressure is below 10 lbs. (0.70 kg/cm²). This is the normal range, but some applications may require a different permissive set point due to specifics of that machine; ie, across-the line starts at full speed. The prelube pump should be sized for the minimum recommended flow rates listed below for each frame model.

Recommended minimum prelube capacity @35 psi (2.46 kg/cm²) (on electric motor driven machines only):

Minimum Recommended Flow Rates

Model	2 Throw GPM (LPM)	4 Throw GPM (LPM)	6 Throw GPM (LPM)
MOS	16 (61)	32 (121)	48 (182)
HOS	15 (57)	25 (95)	50 (189)
HOSS	32 (121)	47 (178)	63 (238)

5-4.1. Hand Pre-Lube Pumps

If a hand pre-lube pump is installed, it should have a flapper type check valve installed in the outlet.

5-5. Oil Parameter Recommendations

Lubricating Oil:

Normal Oil Pressure (at Header)	50-65 PSIG (3.5-31.5 kg/cm ²)
Minimum Oil Pressure (at Header)	40 PSIG (2.8 kg/cm ²)
Shutdown Oil Pressure (at Header)	35 PSIG (2.5 kg/cm ²)
Minimum Start-Up Oil Temperature (Electric Drives)	80°F (27°C)
Minimum Load Oil Temperature (Engine and Electric Drives)	90°F (32°C)
Normal Oil Operating Temperature	150 to 170°F (66 to 77°C)
**NOTE: Not to exceed 180°F (82°C) for continuous operation	